

Double Descent in Linear Models

Samuele Samonini
Student ID: 73427A

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1 Introduction

The central goal of supervised learning is to find a predictor that performs well on unseen data. In the standard framework of empirical risk minimization (ERM), a predictor is chosen from a function class by minimizing the average loss on a training sample, in the hope that a low training error translates into a low error on the underlying data distribution. How to control the capacity of the function class is therefore a fundamental modelling decision.

Classical statistical learning theory answers this question through the *bias-variance trade-off*. Increasing model complexity reduces the bias, since a richer class can represent more of the underlying structure, but it increases the variance, since the model becomes more sensitive to the particular training sample. The result is the familiar U-shaped test-risk curve: error first decreases as complexity grows, reaches a minimum at a “sweet spot” that balances under- and over-fitting, and then increases again. A textbook corollary of this picture is that a model achieving zero training error has overfit the data and should generalize poorly.

Modern machine learning practice openly contradicts this corollary. Very large models, such as deep neural networks, are routinely trained until they exactly fit (interpolate) the training data, and yet they often achieve excellent test performance. Belkin et al. [1] reconcile the classical theory and the modern practice within a single *double descent* curve. Their curve contains the classical U-shape up to the point where the model first interpolates the data, the *interpolation threshold*, but extends it beyond that point: increasing the capacity further makes the test risk descend a second time, frequently reaching values below the classical sweet spot.

The aim of this project is to reproduce the double descent phenomenon empirically in the simplest possible setting, namely linear regression solved by least squares. This setting is attractive for two reasons. First, the phenomenon can be observed transparently: the interpolation threshold has an exact location, and the estimators admit closed-form expressions that are easy to implement from scratch and to reason about. Second, it isolates the mechanism of double descent from the optimization difficulties of neural networks. Following the assignment, we keep the number of training points n fixed and vary the number of features d , so that the interpolation threshold is crossed exactly at $d = n$. We measure training and test error of both ordinary

least squares and ridge regression, and we additionally study the connection between gradient descent and the minimum-norm solution, as well as the effect of label noise on the height of the interpolation peak.

The remainder of this report is organized as follows. Section 2 reviews the theoretical background and the mechanism behind double descent. Section 3 describes the experimental setup: the synthetic dataset, the from-scratch implementation of the estimators, and the evaluation protocol. Section 4 presents and discusses the main results. Section 5 reports the two extensions. Section 6 discusses the positive and negative aspects of the work, and Section 7 concludes.

2 Theoretical Background

We consider linear regression with the squared loss. Given a training set of n examples with inputs in $\mathbb{R}^d$, collected as rows of a matrix $X \in \mathbb{R}^{n \times d}$ and targets $y \in \mathbb{R}^n$, we look for a weight vector $w \in \mathbb{R}^d$ that minimizes the empirical risk

$$\hat{L}(w) = \frac{1}{n} \|Xw - y\|_2^2. \quad (1)$$

The model complexity is identified with the number of features d , while the sample size n is held fixed. As d grows, the behaviour of the least-squares solution changes qualitatively, and it is convenient to distinguish three regimes separated by the *interpolation threshold* $d = n$.

Under-parameterized regime ($d < n$). The system $Xw = y$ is over-determined and has no exact solution in general. Minimizing (1) yields the ordinary least-squares estimator, which is unique whenever X has full column rank:

$$w = (X^\top X)^{-1} X^\top y. \quad (2)$$

Here the classical bias-variance trade-off applies: increasing d reduces the bias but increases the variance, producing the left branch of the U-shaped curve. The training error is strictly positive, because the model cannot fit all n points exactly with fewer than n free parameters.

Interpolation threshold ($d = n$). When $d = n$, the matrix X is (generically) square and invertible, so there is a unique weight vector that fits the training data exactly, giving zero training error. This solution, however, is forced to reproduce every training label, noise included, using exactly as many parameters as there are constraints. It has no freedom left to remain well behaved, and as a result its norm $\|w\|$ typically becomes very large and the test error peaks. The mechanism is most clearly seen through the singular value decomposition $X = U\Sigma V^\top$. The least-squares solution can be written as

$$w = \sum_i \frac{u_i^\top y}{\sigma_i} v_i, \quad (3)$$

where σ_i are the singular values of X . Near the threshold the smallest singular value $\sigma_{\min}$ approaches zero, so the corresponding term in (3) is divided by a vanishing quantity and the solution norm diverges. This is the analytical reason for the spike in both $\|w\|$ and the test error.

Over-parameterized regime ($d > n$). When $d > n$ the system $Xw = y$ is under-determined: there are infinitely many weight vectors that interpolate the training data exactly. Among them, we select the one of *minimum ℓ_2 norm*, which can be written in closed form as

$$w = X^\top (XX^\top)^{-1} y. \quad (4)$$

This is the key choice that produces the second descent. The capacity of the function class alone does not determine how well the predictor generalizes; what matters is the *inductive bias* used to choose among the many interpolating solutions. Selecting the smallest-norm interpolant is a form of Occam’s razor: it favours the simplest explanation compatible with the data. As d increases beyond n , the additional features give more room to satisfy the interpolation constraints with a smaller norm, so $\|w\|$ decreases and the test error descends a second time, often below the classical sweet spot.

Regularization. Ridge regression adds an ℓ_2 penalty to the empirical risk, minimizing $\frac{1}{n}\|Xw - y\|_2^2 + \lambda\|w\|_2^2$ with $\lambda > 0$, which gives the closed-form solution

$$w = (X^\top X + \lambda I)^{-1} X^\top y. \quad (5)$$

The penalty makes the matrix $X^\top X + \lambda I$ invertible for every value of d , so the solution is always unique and its norm is controlled. In effect, regularization prevents the solution from interpolating the noise and bounds the contribution of the small singular values in (3). As we will see, a sufficiently large λ removes the interpolation peak entirely, which also explains why double descent was historically overlooked: any non-trivial regularization, including early stopping, tends to mask it.

Relation to the original paper. Belkin et al. [1] demonstrate double descent for a range of models, including random Fourier features and neural networks. Their Appendix C.4 also analyses a purely linear synthetic model that is conceptually identical to ours, in which ordinary least squares is performed over feature spaces of increasing dimension and the minimum-norm interpolant is selected beyond the threshold. Our experiments can be seen as a minimal and fully transparent instance of that construction.

3 Experimental Setup

3.1 Dataset

The assignment does not prescribe a specific dataset, so we generate a synthetic regression dataset with noisy linear labels. A synthetic design is the natural choice here: double descent is a statement about how the test risk depends on model complexity, and a controlled data-generating process lets us fix the sample size, place the interpolation threshold at a known location, and isolate the phenomenon from confounding factors such as feature correlations or preprocessing artifacts.

We fix an ambient dimension D , which is the maximum number of available features, and sample a true coefficient vector $\beta \in \mathbb{R}^D$ from a standard normal distribution, normalized to unit norm. The inputs are drawn with independent standard Gaussian entries, $x \sim \mathcal{N}(0, I_D)$, and the labels follow the noisy linear model

$$y = x^\top \beta + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2). \quad (6)$$

Normalizing β to unit norm fixes the signal variance to 1, so that the signal-to-noise ratio is controlled solely by σ .

Model complexity is varied through a *subset-of-features* scheme: although the data are generated in the full ambient dimension D , a model of complexity d is fit using only the first d columns of the input matrix. This single mechanism lets us sweep d from 1 to D and cross the interpolation threshold exactly at $d = n$, where n is the (fixed) number of training points. The features that are present in the data but not used by the model act as an additional, structural source of error (model misspecification), which, together with the label noise, drives the peak at the

threshold. This scheme is the linear, fully transparent counterpart of the synthetic construction in Appendix C.4 of Belkin et al. [1].

Training and test sets are generated independently from the same distribution. The model is fit exclusively on the training set, and all reported errors are computed on a separate test set, so no data leakage can occur. We deliberately use a small training set and a large test set: a small n keeps the interpolation threshold at a low, easily reachable complexity, while a large test set yields a low-variance estimate of the test error. The parameters used throughout the main experiment are summarized in Table 1.

Table 1: Parameters of the main experiment.

Parameter	Value	Meaning
D	200	ambient dimension (maximum number of features)
n	40	training set size (fixed); threshold at $d = n$
n_{test}	1000	test set size
σ	0.5	label noise standard deviation
trials	50	independent datasets averaged per complexity
λ	10^{-2}	ridge penalty (baseline value)
seed	0	global random seed (reproducibility)

3.2 Implementation of the estimators

All estimators are implemented from scratch in Python, using `numpy` only for the underlying matrix algebra; no machine-learning library is used to fit the models.

Least squares. Rather than switching between the two closed forms (2) and (4) depending on the regime, we implement a single estimator based on the Moore–Penrose pseudoinverse computed from the singular value decomposition $X = U\Sigma V^\top$. Writing

$$w = V\Sigma^+U^\top y, \quad (\Sigma^+)_{ii} = \begin{cases} 1/\sigma_i & \text{if } \sigma_i > 0 \\ 0 & \text{if } \sigma_i = 0 \end{cases} \quad (7)$$

this one formula returns the ordinary least-squares solution when $d \leq n$ and the minimum-norm interpolating solution when $d > n$, exactly matching the two regimes described in Section 2. Only singular values that are numerically zero (below machine precision) are excluded from the inversion, to avoid division by zero. Crucially, small but non-zero singular values are *kept*: near the threshold it is precisely the smallest singular value that makes the norm blow up, so discarding it would amount to an implicit truncated regularization and would suppress the very peak we want to observe.

Ridge regression. Ridge is implemented directly from its closed form (5), solving the linear system $(X^\top X + \lambda I)w = X^\top y$. The penalty guarantees that the system is always well posed, including in the over-parameterized regime.

Gradient descent. For the extension of Section 5, we also implement batch gradient descent on the squared loss (1), starting from $w = 0$, with the update $w \leftarrow w - \eta \frac{2}{n} X^\top (Xw - y)$. The step size η is set from the largest singular value of X , which bounds the largest eigenvalue of the Hessian and guarantees a stable iteration without manual tuning.

3.3 Evaluation protocol

For each complexity d we record three quantities: the training mean squared error, the test mean squared error, and the norm $\|w\|$ of the learned weight vector. Because each dataset is random, every quantity is averaged over 50 independent trials (a fresh β , inputs and noise per trial) to obtain smooth, representative curves.

The complexities d are sampled on a grid that is intentionally denser around the interpolation threshold $d = n$, where the curves change most sharply, and coarser far from it, where they vary slowly. This keeps the total computation small while resolving the peak accurately. The whole experiment runs in a few seconds on a standard laptop, consistent with the assignment’s emphasis on small- to medium-scale problems and reproducibility rather than training time.

4 Results

We now present the main experimental results. Throughout, the test error is shown on a logarithmic scale, since the peak at the interpolation threshold spans several orders of magnitude. The vertical dashed line marks the interpolation threshold $d = n = 40$.

4.1 The double descent curve

Figure 1 shows the training and test error of ordinary least squares as a function of the model complexity d . The test error exhibits the characteristic double descent shape. In the under-parameterized regime ($d < n$) it follows the classical U-shape, decreasing to a local minimum and then rising sharply as d approaches the threshold. At $d = n$ it reaches a pronounced peak of several orders of magnitude. In the over-parameterized regime ($d > n$) it descends a second time and keeps decreasing as more features are added.

The training error tells the complementary part of the story. It decreases steadily in the under-parameterized regime and collapses to (numerical) zero exactly at the threshold, remaining at zero for all $d > n$. This confirms that every model to the right of the threshold interpolates the training data: they all achieve zero training error, yet their test errors differ dramatically, which is precisely the puzzle that the classical bias–variance picture cannot explain.

A particularly important observation is that the right end of the test-error curve lies *below* the minimum of the classical U-shape on the left. In other words, the heavily over-parameterized model generalizes better than the best under-parameterized model. This is the signature of double descent and the central message of Belkin et al. [1]: increasing capacity beyond the interpolation point can improve generalization. Table 2 reports representative values of the three quantities in the three regimes, summarizing the phenomenon in a few numbers.

Table 2: Representative values (single run) at three complexities: below the threshold, at the threshold, and above it.

	$d = 20$	$d = 40 (= n)$	$d = 80$
Train MSE	5.0×10^{-1}	≈ 0	≈ 0
Test MSE	1.89	132.04	1.59
$\ w\ $	0.93	11.46	0.81

The table makes the three regimes concrete. At $d = 20$ the model does not interpolate (positive train error) and generalizes reasonably. At the threshold $d = 40$ it interpolates (train error ≈ 0) but the test error and the weight norm explode. At $d = 80$ the model still interpolates, yet both the test error and the norm fall back to small values, with the test error even slightly below its value at $d = 20$.

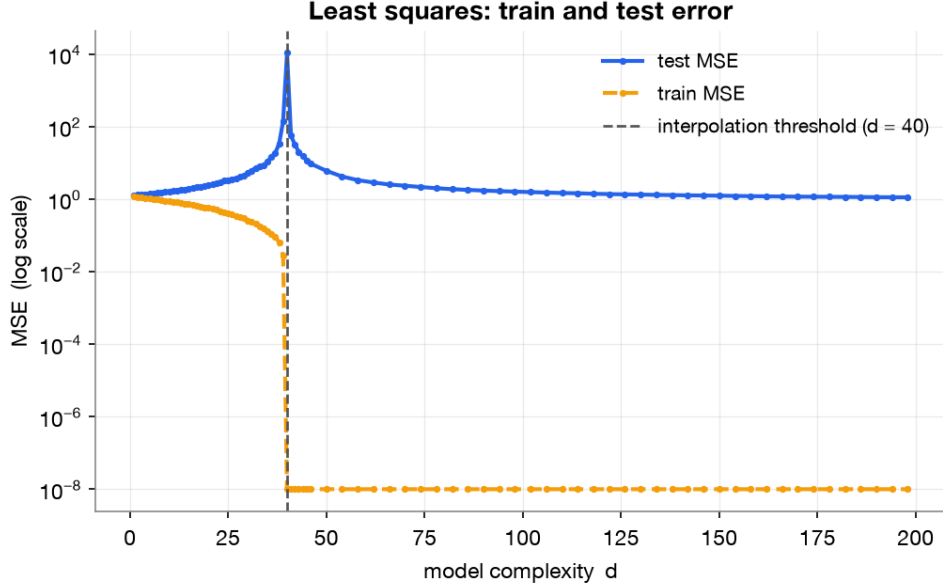


Figure 1: Training and test MSE of least squares versus model complexity d . The test error follows the double descent curve, while the training error drops to zero at the interpolation threshold $d = n$ and stays there. (The training curve is floored at 10^{-8} for display, since past the threshold it reaches machine zero.)

4.2 The solution norm

To understand *why* the test error behaves this way, Figure 2 plots the norm $\|w\|$ of the learned solution against d . The norm peaks exactly at the interpolation threshold and decreases monotonically on both sides. This is the mechanism anticipated in Section 2: at the threshold the smallest singular value of X approaches zero and the solution norm diverges, as predicted by (3); beyond the threshold, the extra features allow interpolating the data with progressively smaller norm. The striking similarity between the shape of the norm curve and that of the test error confirms that the small-norm inductive bias is what governs generalization in this setting.

4.3 Comparison with ridge regression

Finally, Figure 3 compares unregularized least squares with ridge regression for several values of the penalty λ . As λ increases, the interpolation peak is progressively attenuated and, for $\lambda = 1$, removed altogether, leaving only a mild bump. The penalty prevents the solution from interpolating the noise and bounds the contribution of the small singular values, so the model can no longer blow up at the threshold. Far from the threshold, in the over-parameterized regime, all curves converge: there regularization matters little, because the unregularized minimum-norm solution already has small norm. This is exactly the behaviour expected from the theory, and it illustrates why any non-trivial regularization tends to mask double descent.

5 Extensions

This section reports the two extensions suggested by the assignment: the relationship between gradient descent and the closed-form solution, and the effect of the label noise on the interpolation peak.

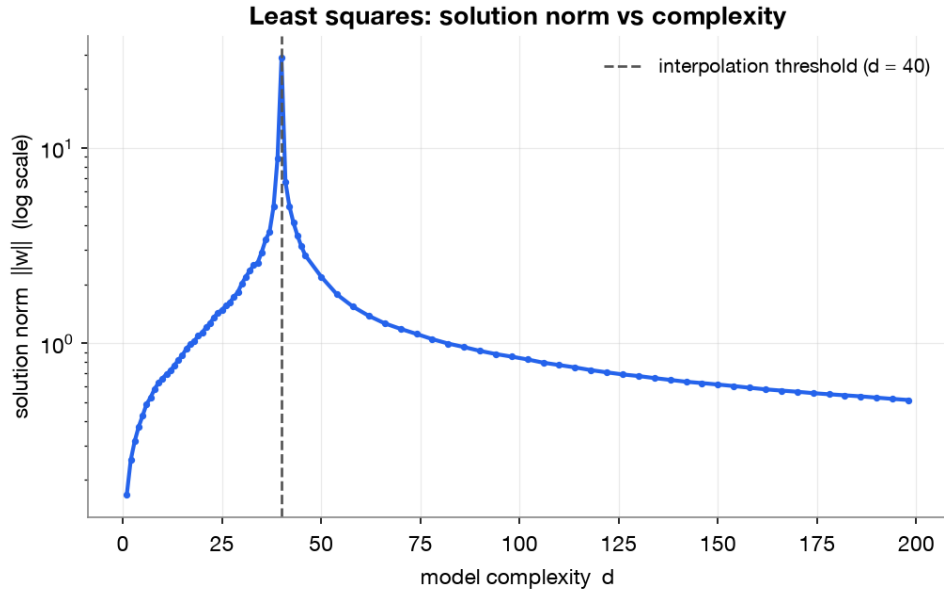


Figure 2: Norm $\|w\|$ of the least-squares solution versus complexity d . The norm peaks at the interpolation threshold and decreases on both sides, mirroring the test-error curve.

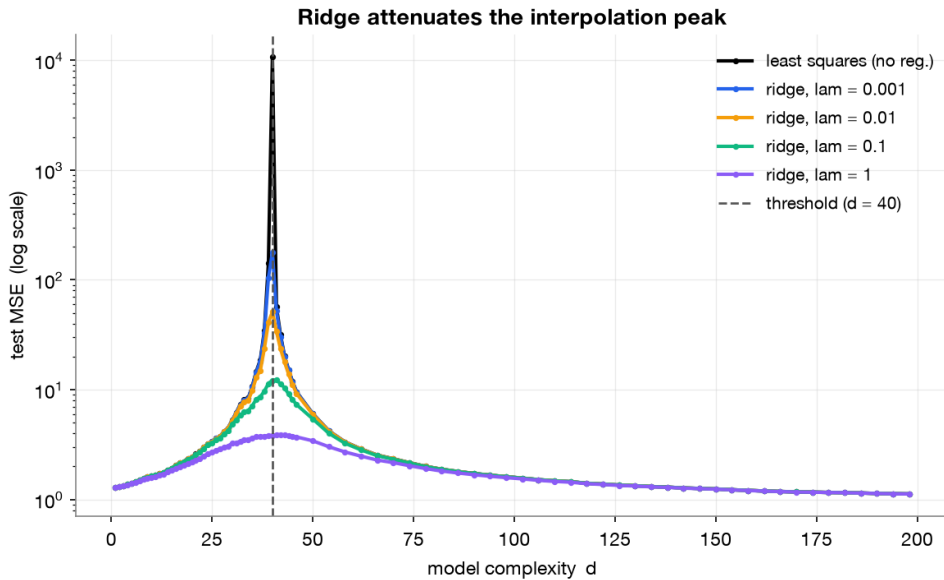


Figure 3: Test MSE of least squares (no regularization) compared with ridge regression for several penalties λ . Increasing λ attenuates and eventually removes the interpolation peak.

5.1 Gradient descent versus closed form

In Section 2 the minimum-norm solution was singled out, among the infinitely many interpolants, by an explicit choice. A natural question is whether a generic optimization algorithm reaches the same solution without being told to minimize the norm. We answer this by running batch gradient descent on the squared loss, started from $w = 0$, and comparing its solution with the closed-form minimum-norm solution.

At a single over-parameterized complexity ($d = 80$), the two solutions are indistinguishable: the difference between the two weight vectors has norm $\approx 2.4 \times 10^{-15}$, i.e. zero up to machine precision, and the two test errors coincide to four decimal places. This is a clean instance of *implicit regularization*: gradient descent from zero initialization, with no penalty whatsoever, selects exactly the low-norm interpolating solution. The same mechanism is invoked by Belkin et al. [1] to explain the inductive bias of the training algorithms used for neural networks.

Figure 4 extends the comparison to the whole grid of complexities. The two test-error curves overlap almost everywhere, so gradient descent reproduces the double descent curve. The only visible deviation is at the threshold, where the gradient-descent peak is slightly lower than the closed-form one. This is itself an instructive effect: near the threshold the problem is ill-conditioned and gradient descent converges very slowly, so stopping after a fixed number of iterations acts as a mild early-stopping regularization. Just like ridge, this small amount of regularization attenuates the peak.

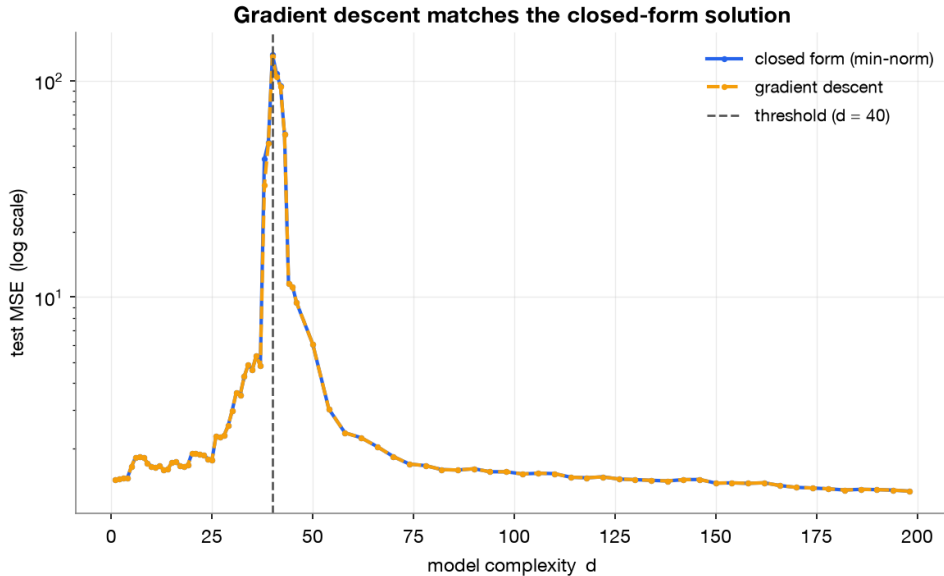


Figure 4: Test MSE of the closed-form minimum-norm solution and of gradient descent from zero initialization, versus complexity d . The two curves coincide almost everywhere; near the threshold the fixed iteration budget acts as a mild early-stopping regularization and slightly lowers the peak.

5.2 Effect of noise

The peak at the interpolation threshold is intimately related to the model fitting the noise exactly. To make this explicit, we repeat the least-squares experiment for several values of the noise standard deviation σ . Figure 5 shows that the height of the peak grows with the noise level: more noise means a larger residual that the interpolating solution is forced to reproduce, which the small singular values then amplify into a taller spike.

Two further observations are worth making. First, even at zero noise ($\sigma = 0$) a residual peak

remains. This peak is not caused by label noise but by model misspecification: when fewer than D features are used, the information carried by the discarded features behaves like an additional source of error that the model cannot fit, and which is amplified at the threshold. Second, far from the threshold the curves are ordered consistently by noise level, but the gap narrows, indicating that the over-parameterized regime is comparatively more robust to noise than the region around the threshold.

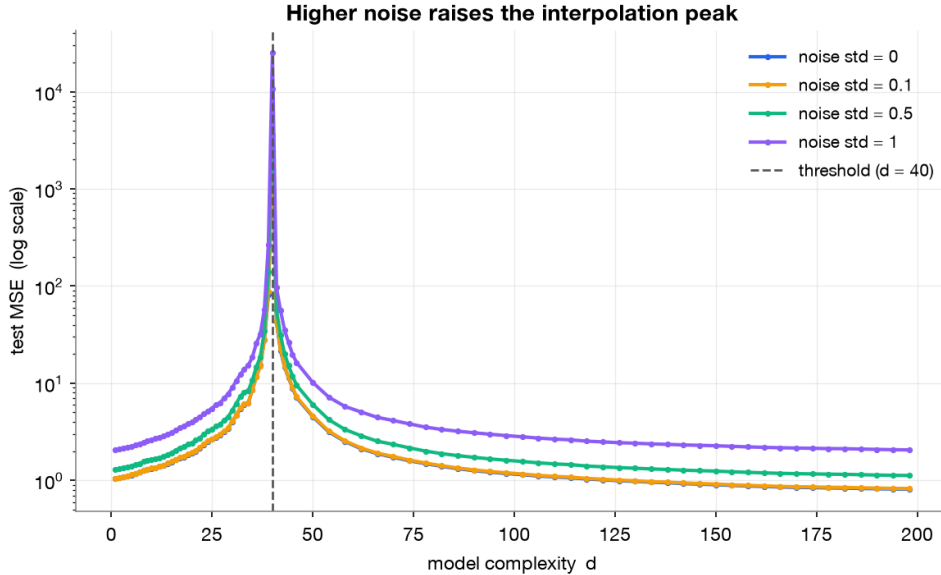


Figure 5: Test MSE of least squares for several label-noise levels σ . The interpolation peak grows with the noise; a smaller residual peak survives even at $\sigma = 0$, caused by model misspecification rather than noise.

6 Discussion

We conclude the analysis by discussing the strengths and the limitations of the work, as required by the assignment.

Positive aspects. The experiments reproduce the double descent phenomenon cleanly and in full. The interpolation peak appears exactly at the predicted location $d = n$, the classical U-shape is recovered in the under-parameterized regime, and the second descent is clearly visible beyond the threshold. Most importantly, the right end of the curve falls below the classical sweet spot, which is the defining feature of double descent and is often the hardest part to obtain in practice. The results also agree quantitatively with the mechanism proposed in the theory: the solution norm $\|w\|$ peaks at the threshold and mirrors the test-error curve, directly linking the phenomenon to the small-norm inductive bias rather than treating it as a black-box observation. The ridge comparison and the two extensions reinforce this reading from independent angles, and the gradient-descent result in particular provides a numerically exact ($\approx 10^{-15}$) confirmation of implicit regularization. Finally, the whole pipeline is implemented from scratch, is fully reproducible from a fixed seed, and runs in a few seconds, which makes every claim easy to verify.

Negative aspects and limitations. Several limitations should be stated honestly. First, the study relies on a single synthetic dataset. This is a deliberate choice that maximizes transparency and control, but it means the conclusions are not tested on real-world data, where feature correlations and distribution shift could change the picture. Second, the precise shape of the curve is

sensitive to the data-generating process: the magnitude of the peak and the depth of the second descent depend on the noise level σ , on the structure of the true coefficient vector β , and on the ratio between the ambient dimension D and the sample size n . We observed, for instance, that the residual peak at zero noise is entirely an artifact of model misspecification, so part of the phenomenon in our setup is driven by the unused features rather than by noise alone. Third, the experiments are confined to linear models; while this is ideal for isolating the mechanism, it does not by itself establish that the same behaviour transfers to the non-linear, non-convex settings of modern deep learning, which we can only motivate by analogy. Finally, the minimum-norm solution near the threshold is numerically delicate: the smallest singular values are close to zero, and although our SVD-based implementation handles this correctly, the exact height of the peak is inherently unstable and should be read qualitatively rather than as a precise number.

7 Conclusion

We have empirically reproduced the double descent risk curve in the simplest possible setting, linear regression with a least-squares estimator, following the construction of Belkin et al. [1]. By fixing the sample size and varying the number of features, we recovered the classical U-shaped curve below the interpolation threshold and a second descent above it, with the over-parameterized regime ultimately outperforming the classical sweet spot. We traced the phenomenon to its underlying mechanism, the explosion and subsequent decay of the solution norm, and we showed that ridge regularization attenuates the peak, that gradient descent from zero initialization recovers the minimum-norm solution through implicit regularization, and that the height of the peak grows with the label noise.

Beyond reproducing a specific curve, the exercise delineates the limits of the classical bias–variance analysis and clarifies what replaces it: in the over-parameterized regime, generalization is governed not by the raw capacity of the model but by the inductive bias used to select among the many interpolating solutions. This perspective is what connects our minimal linear experiment to the behaviour of large neural networks, which are trained well past the point of interpolation and nonetheless generalize well.

References

- [1] M. Belkin, D. Hsu, S. Ma, and S. Mandal. *Reconciling modern machine learning practice and the classical bias–variance trade-off*. Proceedings of the National Academy of Sciences (PNAS), 116(32):15849–15854, 2019.